

CSA0920 - JAVA PROGRAMMING
CO2 Assessment - Question 1 (Direct-Descriptive)

1. HEADER

Name: GOKUL.K

Roll No.: 192521083

Course code & CO reference: CSA0920, CO2

Assignment set: Direct-Descriptive | Question number assigned: Q1

Assigned Question (as given):

Common Runtime Exceptions Showcase

Write a Java program structured into separate methods, each one deliberately triggering and catching a different common runtime exception: `ArithmeticException` (division by zero), `NullPointerException` (calling a method on a null object reference), `ArrayIndexOutOfBoundsException` (invalid array access), and `NumberFormatException` (invalid string-to-number conversion). Display a clear, labeled message for each exception demonstrated.

Concepts: common built-in runtime exceptions.

GitHub repository link: <https://github.com/192521083simats-hue/CSA0920-JAVA-PROGRAMMING.git>

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2. PROBLEM UNDERSTANDING

The task is to demonstrate how common runtime exceptions occur in Java and how they can be handled using try-catch blocks. Each exception is triggered intentionally in a separate method so that the program remains structured and easy to understand. The program shows how errors like division by zero, null object access, invalid array index, and incorrect number format can be safely handled without crashing the program.

3. APPROACH / DESIGN NOTES

1. Used a single Main class with multiple methods, each dedicated to one exception.
2. Each method contains a try-catch block to handle the specific exception.
3. `ArithmeticException` is triggered using division by zero
4. `NullPointerException` is triggered by calling a method on a null object.
5. `ArrayIndexOutOfBoundsException` is triggered by accessing an invalid array index.
6. `NumberFormatException` is triggered by converting a non-numeric string to integer.

7. Clear messages are printed for each exception to make output understandable.

4. SOURCE CODE

```
public class Main {

    public static void main(String[] args) {
        handleArithmeticException();
        handleNullPointerException();
        handleArrayIndexException();
        handleNumberFormatException();
    }

    // ArithmeticException
    public static void handleArithmeticException() {
        try {
            int a = 10 / 0;
        } catch (ArithmeticException e) {
            System.out.println("ArithmeticException: Cannot divide by zero.");
        }
    }

    // NullPointerException
    public static void handleNullPointerException() {
        try {
            String str = null;
            System.out.println(str.length());
        } catch (NullPointerException e) {
            System.out.println("NullPointerException: Cannot call method on null object.");
        }
    }

    // ArrayIndexOutOfBoundsException
    public static void handleArrayIndexException() {
        try {
            int arr[] = {1, 2, 3};
            System.out.println(arr[5]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("ArrayIndexOutOfBoundsException: Invalid array index accessed.");
        }
    }

    // NumberFormatException
    public static void handleNumberFormatException() {
        try {
            int num = Integer.parseInt("abc");
        } catch (NumberFormatException e) {
            System.out.println("NumberFormatException: Invalid string for number conversion.");
        }
    }
}
```

5. TEST CASES

1.Input:Division by zero

Expected Output:ArithmeticException message

Actual Output:ArithmeticException: Cannot divide by zero

2.Input:Null string

Expected Output:NullPointerException message

Actual Output:NullPointerException: Cannot call method on null object

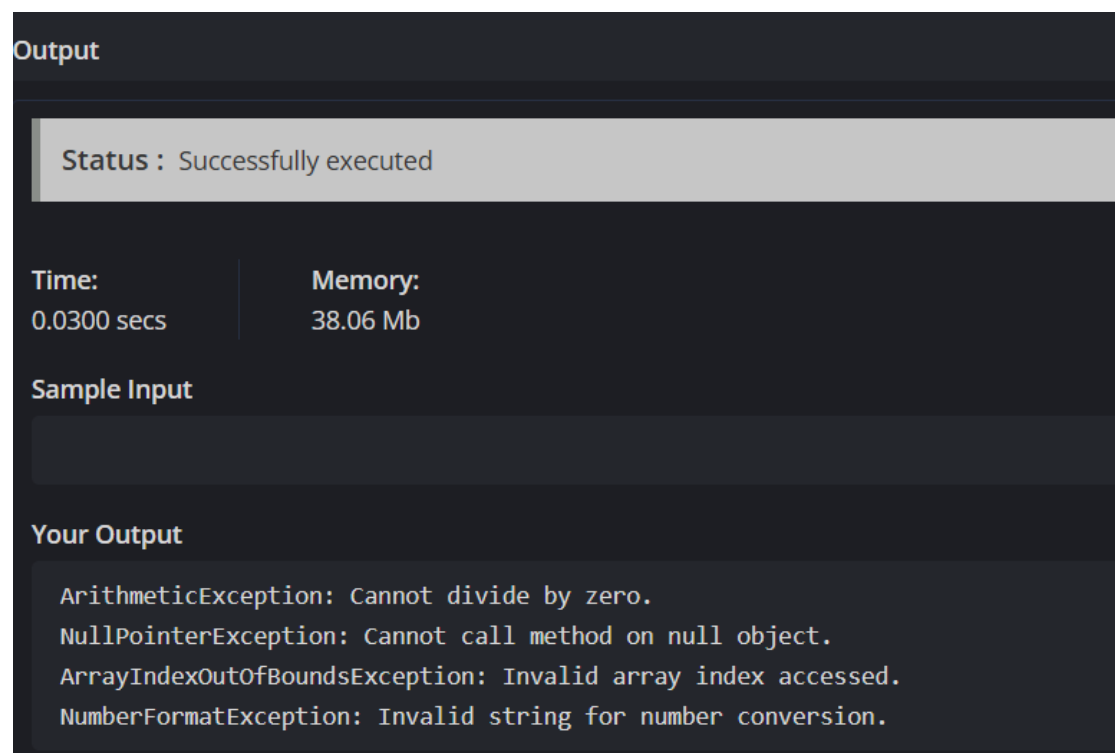
3.Input:Invalid array index

Expected Output:ArrayIndexOutOfBoundsException message

Actual Output:ArrayIndexOutOfBoundsException: Invalid array index accessed

6. SCREENSHOTS OF EXECUTION

Screenshot 1 - ArithmeticException,NullPointerException, ArrayIndexOutOfBoundsException, NumberFormatException.



7. REFLECTION

This program helped in understanding how runtime exceptions occur during program execution and how they can be handled gracefully using try-catch blocks. Handling exceptions properly prevents program crashes and improves reliability. Each exception demonstrated a common mistake that programmers may encounter, making this exercise practical and useful.

8. DECLARATION

"This is my own work. Any AI/external tool assistance used is disclosed above."